

RESEARCH ARTICLE

Semi-Supervised Learning Under Extreme Class Imbalance in Wafer Bin Map Defect Pattern Classification

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ABSTRACT The semiconductor industry faces significant challenges in wafer quality assessment due to the increasing complexity of device fabrication and the rise in inline defects and circuit failures. As manufacturing processes advance, conventional manual analysis of wafer bin maps (WBMs) has become impractical and unreliable, unable to keep pace with the industry's demands. While convolutional neural networks (CNNs) are widely used in automated systems for classifying WBM defect patterns, their dependence on supervised learning limits their effectiveness, as they cannot utilize the vast amounts of unlabeled WBM data. Additionally, class imbalances in WBM datasets often result in models biased toward majority classes, leading to suboptimal performance. In this study, we produce ClimEx, a semi-supervised learning framework that addresses the extreme class imbalance problem commonly found in WBM datasets through consistency regularization and pseudo-labeling. Our method maintains dynamic per-class thresholds to correct the imbalance in pseudo-labels caused by biased predictions, ensuring balanced representation across classes even under extreme imbalance. On a publicly available benchmark, ClimEx demonstrates improved predictive performance compared to state-of-the-art methods. We also provide a comprehensive analysis of the key components of our method from multiple perspectives.

INDEX TERMS Wafer bin map defect pattern classification, semiconductor manufacturing, semi-supervised learning, class-imbalanced learning, class-imbalanced semi-supervised learning.

I. INTRODUCTION

The semiconductor industry is continuously striving to increase its productivity and satisfy the increasing demands of the global market. However, the intricate process of semiconductor device fabrication, compounded by decreasing node sizes and the integration of more chips on a single wafer, has led to a rise in inline defects and circuit failures. Such issues adversely affect the overall yield, prompting manufacturers to seek advanced techniques for wafer

quality assessment to reduce production costs and sustain competitiveness.

The analysis of spatial defect patterns in wafer bin maps (WBMs) is an area of significant interest. WBMs are graphical representations of the test results for individual chips on a semiconductor wafer, indicating whether each chip passed or failed specific quality tests. The spatial pattern of defects provides crucial insights into process failures and their underlying causes and has traditionally been analyzed manually by experienced engineers. However, with modern fabs producing thousands of wafers per week, this manual approach is becoming untenable and unreliable. Furthermore, the vast amounts of unexamined WBM data represent a

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missed opportunity to identify root causes and improve yield.

To address these challenges, automated systems capable of effectively analyzing WBM defect patterns are essential. Recently, convolutional neural networks (CNNs) have emerged as a popular and standard choice for classifying WBM defect patterns [1], [2]. However, these methods are highly dependent on supervised learning with labeled data, which are often limited in quantity and potentially biased due to the subjective nature of expert domain involvement in the labeling process [3].

Meanwhile, in the field of computer vision, the use of unlabeled data through semi-supervised learning (SSL) has recently gained momentum, particularly in situations where labeled data are scarce [4], [5], [6], [7], [8], [9], [10]. Several recent studies have also explored the use of unlabeled data to train models to classify WBM defect patterns [11], [12], [13], [14]. However, these studies often overlook the issue of class imbalance in WBMs, where the distribution of samples between different classes is uneven [3]. Unlike relatively balanced scenarios commonly seen in computer vision benchmarks [15], [16], [17], imbalanced distributions of defect patterns are a prevalent challenge that has not yet been adequately addressed. This imbalance can significantly skew the model predictions, affecting the overall classification performance and especially disadvantaging underrepresented classes [18].

Building on recent advances in class-imbalanced semi-supervised learning (CISSL) [19], [20], [21], we propose ClimEx, a novel framework specifically designed for WBM defect pattern classification under extreme class imbalance. While existing methods like Adsh [21] employ dynamic thresholding to mitigate bias, they often suffer from severe threshold fluctuations in extremely imbalanced scenarios, leading to training instability. ClimEx addresses this limitation by integrating consistency regularization, pseudo-labeling, and a momentum-based dynamic thresholding mechanism. Specifically, the momentum update stabilizes the volatile dynamic thresholds required for handling the extreme imbalance (approx. 85% majority) characteristic of WBM data. By ensuring a balanced representation across classes without collapsing into trivial solutions, our method demonstrates superior stability and predictive performance compared to state-of-the-art algorithms on a publicly available benchmark.

The remaining sections of this paper are organized as follows: Section II provides a brief review of related work on WBM defect pattern classification and recent advances in class-imbalanced semi-supervised learning. The details of the proposed methodology is illustrated in Section III. The experimental results are reported in Section IV. Section V provides a discussion on threshold fluctuation and its stabilization, highlighting the significance of these aspects in our research. Finally, Section VI presents concluding remarks and outlines for future research.

II. RELATED WORK

A. WAFER BIN MAP PATTERN CLASSIFICATION

WBM defect pattern classification is a crucial task in the semiconductor industry, playing a vital role in quality control and yield improvement. However, this task faces significant challenges, including the need for large amounts of labeled data and the inherent class imbalance in defect patterns. These challenges have led to extensive studies exploring various approaches to improve classification accuracy and efficiency.

CNNs have gained considerable attention for WBM defect pattern classification due to their remarkable performance in image classification by preserving spatial information. One study focused on the classification of mixed-type defect patterns using CNNs, proposing an architecture designed to capture both local and global features of WBMs [1]. Another investigation explored CNN-based classification of defect patterns using a synthetic WBM dataset and demonstrated their applicability to wafer map image retrieval [22]. Further work addressed class imbalance in WBM datasets through a data augmentation-based oversampling strategy combined with CNNs, highlighting the challenge posed by imbalanced data [23]. In addition, a supervised contrastive learning technique was used to enhance feature representations within a CNN training framework, resulting in improved WBM pattern classification accuracy [2].

Although these CNN-based methods have shown promising results, they largely rely on supervised learning, which demands a substantial amount of labeled data. In an effort to alleviate this requirement, recent research has leveraged unlabeled data, which are abundantly available even in cases where labeled data are scarce. One approach introduced a semi-supervised and incremental modeling framework for WBM classification that incorporates both labeled and unlabeled data, effectively improving classification accuracy under limited labeled data conditions [11]. Another model used pretext-invariant representation learning (PIRL) and data augmentation techniques tailored to WBMs [12], while a different study developed a semi-supervised WBM pattern recognition method integrating domain-specific data augmentation and contrastive learning [13].

These approaches demonstrated effective use of unlabeled data for enhanced classification performance but did not explicitly address class imbalance within their learning frameworks. To resolve this issue, some recent efforts have proposed more sophisticated methods. One such method applied an ensemble-based SSL approach that combines outputs from multiple CNN models, mitigating the impact of class imbalance by leveraging the diversity of the ensemble [24]. Another study introduced a dual-head CNN with two classification heads—one focusing on overall accuracy and the other on class-wise accuracy—thus handling imbalance by encouraging attention to minority classes [14].

However, these methods do not explicitly address class imbalance within both consistency regularization and pseudo-labeling frameworks, often relying on the

architecture's inherent ability to balance class-specific and overall performance. In contrast, our method focuses on improving the quality of pseudo-labels through an adaptive thresholding mechanism, which dynamically updates thresholds to reduce the likelihood of incorrect pseudo-labeling. This approach inherently aligns with consistency regularization and ensures better representation of both majority and minority classes throughout the learning process.

B. CLASS-IMBALANCED SEMI-SUPERVISED LEARNING

We aim to address the challenge of WBM defect pattern classification by using unlabeled WBM data for effective representation learning, while considering the extreme class imbalance present in WBM datasets. SSL aims to enhance the model's performance using unlabeled data, thereby reducing reliance on the quantity of labeled data. Prominent approaches include the concepts of pseudo-labeling and consistency regularization [4], [5], [6], [7], [8], [9], [10].

Pseudo-labeling involves the use of the predictions of a trained model to generate pseudo-labels for unlabeled data, which are subsequently used for further model training in an iterative manner [4]. Consistency regularization aims to enhance the robustness and generalization of a model by leveraging unlabeled data to penalize the prediction variations caused by input perturbations [6]. It can employ either soft [5], [7], [8], [9] or hard [10] pseudo labels in the regularization process.

These studies demonstrated state-of-the-art performance on classic image benchmarks and have approached the performance of supervised learning when all data were labeled. However, in more realistic scenarios, such as the WM-811K dataset with class imbalance, SSL algorithms exhibit poor performance because they are biased toward the majority class [19], [20], [21]. In particular, in SSL algorithms that employ pseudo-labeling and consistency training, an imbalanced class distribution results in biased model predictions in favor of the majority class, resulting in pseudo labels that are also biased toward the majority class. Additionally, the computation of the consistency loss is dominated by the losses associated with the majority class.

Recently, CISSL algorithms have been proposed to enhance the robustness and generalization of SSL algorithms in real-world scenarios characterized by class imbalance. [19] proposed a method to refine raw biased pseudo labels via convex optimization, addressing the challenges posed by class imbalance in SSL. This approach used pseudo-labeling and introduced a novel refinement process to align the distribution of labeled and unlabeled data, resulting in improved performance in class-imbalanced scenarios. Another study introduced class rebalancing self-training, a self-training framework for CISSL that addressed class imbalance by iteratively reweighting and relabeling samples [20]. Their approach achieved better performance by effectively leveraging unlabeled data and rebalancing class distributions during training. However, these studies have

strict assumptions, such as the need for prior knowledge of the class distribution of unlabeled data or the assumption that the class distribution should be identical between labeled and unlabeled data. In the case of WM-811K, the class distribution of the unlabeled data remains unknown.

A more recent work proposed an adaptive thresholding method for CISSL [21]. Their approach dynamically adjusted the decision threshold based on logits or probabilities of predicted classes, effectively using labeled and unlabeled data in imbalanced scenarios. This method improved the performance of SSL algorithms without requiring class distribution constraints.

Our method takes advantage of the adaptive thresholding approach to address the extreme class imbalance in WBM defect pattern classification. We further improve the adaptive thresholding mechanism of [21] by incorporating a momentum-based update strategy. This modification allows for more stable and effective threshold adjustments, particularly in scenarios with severe class imbalance such as those encountered in WBM datasets. By combining the flexibility of adaptive thresholding with the stability of momentum updates, we achieve improved performance in handling the unique challenges posed by WBM classification tasks. Our work not only addresses the limitations of previous CISSL methods, but also pushes the boundaries of classification accuracy in extremely imbalanced, semi-supervised scenarios typical in semiconductor manufacturing processes.

III. METHODOLOGY

In this section, we introduce our CISSL framework for WBM defect pattern classification. We begin with the problem setting in Section III-A, followed by relevant background in Section III-B, and then detail the proposed method in Section III-C. An overview of our method is shown in Figure 1.

A. PROBLEM SETTING

We consider the standard SSL setting where we are presented with a labeled WBM dataset $\mathcal{D}^l = \{(\mathbf{x}_1^l, y_1^l), \dots, (\mathbf{x}_N^l, y_N^l)\}$ and an unlabeled WBM dataset $\mathcal{D}^u = \{\mathbf{x}_1^u, \dots, \mathbf{x}_M^u\}$, where $\mathbf{x} \in \mathcal{X}$ represents the input WBM and $y \in \mathcal{Y} = \{1, 2, \dots, K\}$ represents the labels for a K -way defect pattern classification task. The goal is to learn a probabilistic deep neural network classifier $f : \mathcal{X} \rightarrow \Delta^{K-1}$ on the basis of $\mathcal{D}^l$ and $\mathcal{D}^u$. Given the batch-wise nature of neural network training, let $\mathcal{B}^l = \{(\mathbf{x}_i^l, y_i^l) : i = 1, \dots, B\}$ be a batch of B labeled examples sampled from $\mathcal{D}^l$, and $\mathcal{B}^u = \{\mathbf{x}_i^u : i = 1, \dots, \mu B\}$ be a batch of μB unlabeled examples sampled from $\mathcal{D}^u$. Note that $\mu \in \mathbb{N}$ is merely a hyperparameter that determines the relative batch size of $\mathcal{B}^u$ with respect to $\mathcal{B}^l$. For the purpose of illustration, we assume that the K classes are sorted in descending order according to the sample sizes, starting with the majority class with the largest sample size being class 1, and ending with the minority class with the smallest sample size being class K .

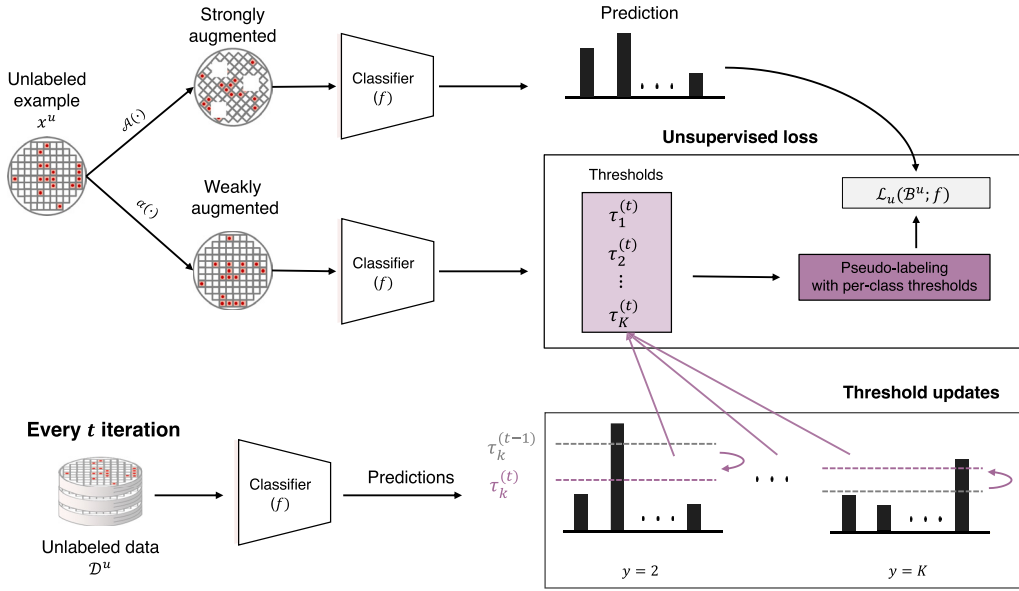


FIGURE 1. A schematic overview of our method, ClimEx, highlighting the process of per-class threshold updates. For simplicity, the supervised learning component is omitted. Each unlabeled example is processed through weak and strong augmentation pipelines: the weakly augmented version generates pseudo-labels, while the strongly augmented version serves as the target for consistency regularization. Per-class thresholds are dynamically updated to ensure balanced pseudo-label generation.

TABLE 1. Comparison of ClimEx with existing semi-supervised learning frameworks. ClimEx is explicitly designed to handle extreme class imbalance in WBM datasets by stabilizing dynamic thresholds.

Feature	FixMatch [10]	Adsh [21]	ClimEx (Ours)
Class-wise threshold	× (Fixed)	✓	✓
Threshold update	Constant	Instant (Batch-wise)	Momentum-based
Stability mechanism	N/A	× (Fluctuation)	✓ (Stabilized)
Focus on extreme imbalance	Low	Partial	Explicitly modeled
Target domain	General vision	Generic imbalance	WBM-specific

B. BACKGROUND

To fully utilize the information in the unlabeled WBM data, our proposed method is based on two key principles of deep SSL: *consistency regularization* and *pseudo-labeling* [10]. Consistency regularization promotes semantic stability in a probabilistic deep neural network classifier f by encouraging it to produce consistent predictions across diversely perturbed (or augmented) versions of the same input. This technique enhances the model's ability to capture invariant patterns, thereby improving generalization, particularly in scenarios with limited labeled data.

Technically, this approach minimizes the discrepancy between the pseudo-labels generated by hard-assigning the probabilistic prediction on one perturbed version of the input, obtained through a weak stochastic augmentation operator $\alpha : \mathcal{X} \rightarrow \mathcal{X}$, and the raw predicted probability distribution from another perturbed version, created via a strong stochastic augmentation operator $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{X}$:

$$\min_f \mathbb{E} \left[d(f(\alpha(\mathbf{x})), f(\mathcal{A}(\mathbf{x}))) \right]$$

where $d : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$ is a functional measure of discrepancy between two probability distributions (e.g., cross entropy). It is to note that the choice of weak and strong augmentations is a critical factor in maximizing the effectiveness of these simple yet powerful techniques. For our WBM defect pattern classification task, we tailored the augmentations by exploring the five WBM augmentations introduced by [12].

Specifically, the *rotation* strategy, where WBMs are geometrically rotated to some random degree, may serve as a form of weak augmentation, aiming to minimally alter input data while preserving essential WBM attributes. Meanwhile, a combination of *cropping*, *cutout*, *noise addition*, and *shifting* can serve as strong augmentations, introducing more pronounced variations that challenge the model to maintain consistent predictions against, thereby promoting robustness and enhancing its ability to generalize from complex data patterns. We empirically justify the choice of our augmentations in Section IV-C.

C. PROPOSED METHOD

The semi-supervised training loss of our proposed method is composed of a supervised loss $\mathcal{L}_s$ based on labeled examples $\mathcal{B}^l$ and an unsupervised loss $\mathcal{L}_u$ based on unlabeled examples $\mathcal{B}^u$. Similar to [10], we formulate the supervised loss $\mathcal{L}_s$ using the standard categorical cross-entropy loss with weakly augmented labeled examples as follows:

$$\mathcal{L}_s(\mathcal{B}^l; f) \triangleq -\frac{1}{B} \sum_{i=1}^B \sum_{k=1}^K \mathbb{1}(y_i^l = k) \cdot \log \left[f(\alpha(\mathbf{x}_i^l)) \right]_k \quad (1)$$

where $\mathbb{1}(\cdot) \in \{0, 1\}$ is the indicator function, and $f(\mathbf{x})$ represents the predicted class probability vector given input $\mathbf{x}$. $[f(\mathbf{x})]_k$ denotes the predicted probability associated with the k -th class.

To construct the unsupervised loss $\mathcal{L}_u$, hard pseudo-labeling is performed by assigning a deterministic label corresponding to the class with the highest predicted probability produced by the model f given a weakly augmented unlabeled example $\alpha(\mathbf{x}_i^u)$. In other words, if the model assigns highest probability to class k , the pseudo-label is equal to k . Let $\mathbf{q}_i^u = f(\alpha(\mathbf{x}_i^u))$ denote the predicted probability vector given the i -th weakly augmented unlabeled example. Mathematically, hard pseudo-labeling can be expressed as follows:

$$\hat{y}_i^u = \arg \max_k [\mathbf{q}_i^u]_k. \quad (2)$$

Once the pseudo-label is obtained using Eq. (2), consistency regularization is introduced to minimize the discrepancy against the probabilistic predictions made on top of the strongly augmented version $\mathcal{A}(\mathbf{x}_i^u)$. This enables the model to capture better generalizable representations of WBMs. The consistency regularization term for the i -th unlabeled example is formulated using the categorical cross entropy function:

$$-\sum_{k=1}^K \mathbb{1}(\hat{y}_i^u = k) \cdot \log [f(\mathcal{A}(\mathbf{x}_i^u))]_k \quad (3)$$

It is worth noting that in a class-balanced scenario where classes are uniformly distributed, we need not consider the per-class quantity of pseudo-labels in a training batch, but only the quality of the pseudo-labels [10]. Thus, it is reasonable to allow unlabeled examples with high maximal predictive confidence to participate in training, given that it exceeds a fixed high threshold τ equally applied to all classes (e.g., $\tau = 0.95$). We may formulate an unsupervised loss function given an unlabeled batch $\mathcal{B}^u$ with a fixed threshold τ as follows:

$$-\frac{1}{\mu B} \sum_{i=1}^{\mu B} \sum_{k=1}^K \mathbb{1}(\max(\mathbf{q}_i^u) \geq \tau) \cdot \mathbb{1}(\hat{y}_i^u = k) \cdot \log [f(\mathcal{A}(\mathbf{x}_i^u))]_k \quad (4)$$

However, in a class-imbalanced scenario which is commonly observed in WBM data [3], using a fixed threshold as for Eq. (4) can result in low recall rates for minority classes [20], [21]. The underlying reason for this bias towards the majority class(es) lies in the skewed class distribution of the labeled training data. With an imbalanced class distribution, the model encounters a higher frequency of examples from the majority classes, resulting in an overemphasis on accurately predicting them. In other words, the model's responsiveness to minority classes diminishes, reducing both recall and accuracy in identifying under-represented classes. This bias impedes the ability of the model

to generalize effectively on the minority classes, resulting in poor predictive performance.

Class imbalance poses a significant challenge for semi-supervised learning, particularly in the early stages of training where the model is prone to generate overly confident predictions for the majority classes [20], [25]. These confident predictions rapidly drive down the training loss, erroneously reinforcing the model's bias towards the majority classes. Consequently, the model struggles not only to classify minority classes correctly, but also to produce sufficiently confident predictions exceeding the threshold τ for these classes. This results in skewed pseudo-labeling, where most of the pseudo-labels are assigned to the majority classes, further exacerbating class imbalance.

To mitigate this problem, we propose to employ a class-dependent threshold instead of a fixed global threshold [21]. Concretely, our proposed method uses class-wise thresholds $\tau_k^{(t)}$ for $k \in \{1, \dots, K\}$ which are dynamically updated at any training iteration $t \in \{1, \dots, T\}$. To simplify the notation, let $\mathbf{p}_i^u = (p_{i1}^u, p_{i2}^u, \dots, p_{iK}^u) = f(\alpha(\mathbf{x}_i^u))$ be the predicted probability vector for the i -th weakly augmented unlabeled example. For any class k at iteration t , we first define a set $\mathcal{C}_k^{(t)}$ whose elements are the probabilities p_{ik}^u for which k is the class index that achieves the maximum probability in the vector $\mathbf{p}_i^u$, for $i \in \{1, \dots, \mu B\}$. In other words, $\mathcal{C}_k^{(t)}$ is the collection of probabilities where class k is the most likely class according to the model's predictions for each unlabeled example. This can be mathematically expressed as:

$$\mathcal{C}_k^{(t)} \triangleq \left\{ p_{ik}^u \mid \arg \max_{j \in \{1, \dots, K\}} p_{ij}^u = k, i \in \{1, \dots, \mu B\} \right\}. \quad (5)$$

Specifically, for each unlabeled example in $\mathcal{D}^u$, the model predicts a probability distribution over all possible classes (e.g., via a softmax). We then look at which class is assigned the highest probability. If class k is associated with the highest probability, that probability is included in $\mathcal{C}_k^{(t)}$. In other words, $\mathcal{C}_k^{(t)}$ gathers all probabilities where class k was the top choice according to the model.

Given $\mathcal{C}_k^{(t)}$, the next step is to determine a per-class probability to use as a threshold $\tau_k^{(t)}$ that allows an approximately equal proportion of unlabeled examples per class to be pseudo-labeled and participate in model training. To achieve this, we first calculate the proportion $\rho^{(t)} \in [0, 1]$ of pseudo-labeled examples which are to belong to the majority class ($k = 1$) using an arbitrary threshold $\tau_1^{(t)}$ (e.g., $\tau_1^{(1)} = \tau_1^{(2)} = \dots = \tau_1^{(T)} = 0.95$):

$$\rho^{(t)} = \frac{\sum_{p \in \mathcal{C}_1^{(t)}} \mathbb{1}(p \geq \tau_1^{(t)})}{|\mathcal{C}_1^{(t)}|}, \quad (6)$$

which is essentially the proportion of unlabeled examples who are to participate in training with a pseudo-label of $y = 1$. Based on $\rho^{(t)}$, we determine the dynamic thresholds $\tau_k^{(t)}$ for $k \in \{2, \dots, K\}$. Let $N(p) = |\{q \in \mathcal{C}_k^{(t)} : q \geq p\}|$ be the

number of candidates with probability at least p . We set:

$$\tau_k^{(t)} = \min \left\{ p \in \mathcal{C}_k^{(t)} \mid \frac{N(p)}{|\mathcal{C}_k^{(t)}|} \geq \rho^{(t)} \right\}. \quad (7)$$

Herein, we seek the smallest probability p such that the proportion of elements in $\mathcal{C}_k^{(t)}$ whose values are greater than or equal to p is at least $\rho^{(t)}$. This approach ensures that pseudo-labels are selected with thresholds reflecting equivalent levels of prediction confidence across all classes, thereby balancing class representation in the pseudo-labeling process.

Using class-specific thresholds dynamically updated over time, the unsupervised loss $\mathcal{L}_u$ is reformulated as follows:

$$\mathcal{L}_u(\mathcal{B}^u; f) = -\frac{1}{\mu B} \sum_{i=1}^{\mu B} \sum_{k=1}^K \mathbb{1}(\max(\mathbf{q}_i^u) > \tau_{y_i^u}^{(t)}) \cdot \mathbb{1}(\hat{y}_i^u = k) \log [f(\mathcal{A}(\mathbf{x}_i^u))]_k. \quad (8)$$

Ultimately, our proposed loss function is defined as the sum of the supervised loss from Eq. (1) and the unsupervised loss from Eq. (8):

$$\mathcal{L}(\mathcal{B}^l, \mathcal{B}^u; f) = \mathcal{L}_s(\mathcal{B}^l; f) + \mathcal{L}_u(\mathcal{B}^u; f) \quad (9)$$

In early-stage evaluations where models were trained based on the thresholds obtained via Eq. (7), we observed unexpected fluctuations in the thresholds over time, which likely contributed to suboptimal classification performance. We attribute this challenge primarily to the extreme class imbalance present in WBM defect patterns. As shown in Table 2, the majority class, ‘None’, comprises 85.24% of the labels, overwhelmingly dominating the other classes. We empirically support this conjecture in Section V.

To address this issue, we propose a momentum-based update for the per-class thresholds as a simple yet effective solution to prevent fluctuations and stabilizing model training:

$$\tau_k^{(t)} \leftarrow (1 - \lambda) \tau_k^{(t-1)} + \lambda \tau_k^{(t-1)}. \quad (10)$$

In Eq. (10), $\lambda \in [0, 1]$ is the momentum coefficient that controls the influence of the previous threshold value on the current update. A higher λ assigns greater weight to the thresholds from the previous iteration. As updates are iteratively applied, the influence of past values accumulates exponentially, moderated by λ , thereby shaping the current threshold. In preliminary experiments, we conducted a grid search with λ values in $[0.1, 0.2, 0.5, 0.8, 0.9]$ and found that $\lambda = 0.8$ provided the best performance in practice.

IV. EXPERIMENTS

A. DATASET

We evaluated the performance of our method on the publicly available benchmark for WBM defect pattern classification, WM-811K [3], which consists of 811,457 WBMs acquired from a real-world fabrication process. Only 172,950 WBMs

TABLE 2. Basic statistics of the WM-811K benchmark.

Class label	Sample size	Proportion (%)
None	147,431	85.24
Edge-Ring	9,680	5.60
Edge-Loc	5,189	3.00
Center	4,294	2.48
Loc	3,593	2.08
Scratch	1,193	0.69
Random	866	0.50
Donut	555	0.32
Near-Full	149	0.09
Unlabeled	Sample size	Proportion (%)
-	638,507	100.00

are associated with human-annotated labels, each representing one of nine defect patterns ($K = 9$). The remaining 638,507 form a significantly large unlabeled set. The class distribution is highly imbalanced, with the dominant class accounting for approximately 85% of the entire population. Detailed sample counts and proportions are presented in Table 2.

Each WBM represents the binary test results of a semiconductor wafer, whose values indicate the presence or absence of defects. Normal dies have a value of one, while defective dies have a value of two. Locations outside the specified circular WBM region are assigned values of zero. Because the spatial dimension varies across WBMs, we resized them to a 96×96 image based on nearest neighbor interpolation following [12].

B. TRAINING CONFIGURATIONS

The labeled dataset was divided into three subsets: 80% for training, 10% for validation, and 10% for testing. Following the standard evaluation protocol used in several SSL studies, [10], [26], [27], [28], we evaluated model performance in scenarios with limited labeled data. Specifically, a random portion of the labeled data, along with the entire unlabeled dataset, is used for training. We considered various scenarios, ranging from 1% (1,383 instances) of the labeled training data to 100% (138,360 instances), allowing us to assess the model’s performance across different labeled data sizes.

We used ResNet-18 [29] as the CNN backbone for feature extraction, with a linear classifier stacked on top, consisting of a fully connected layer followed by a softmax activation function to convert the output logits into valid probabilities. To mitigate potential overfitting, dropout with a probability of 0.5 was applied before the linear classifier [30]. Models were trained for 51,200 iterations with a batch size of 256 for both labeled and unlabeled data (i.e., $\mu = 1$). Pseudo-labeling thresholds $\tau_2^{(t)}, \tau_3^{(t)}, \dots, \tau_9^{(t)}$ were dynamically updated every 512 training iterations, while keeping $\tau_1^{(t)} = 0.95$ constant. We used the Adam optimizer with a base learning rate of 0.001 and default hyperparameters ($\beta_1 = 0.9, \beta_2 = 0.999$) [31]. The base learning rate was selected through a grid search over the values $\{0.01, 0.005, 0.001, 0.0005, 0.00001\}$.

For model selection, we measured model performance on the held-out validation set every 512 iterations using the macro F_1 score, a widely used metric for classification tasks with class imbalance. This metric is computed by first calculating the F_1 score for each class and then taking their macro-average, ensuring equal contribution from each class to the overall score, regardless of class prevalence. The F_1 score for class k is the harmonic mean of precision and recall for that class, defined as follows:

$$F_1(k) = 2 \times \frac{\text{Precision}(k) \times \text{Recall}(k)}{\text{Precision}(k) + \text{Recall}(k)},$$

where $\text{Precision}(k) = \frac{TP(k)}{TP(k) + FP(k)}$ and $\text{Recall}(k) = \frac{TP(k)}{TP(k) + FN(k)}$. Here, $TP(k)$, $FP(k)$, and $FN(k)$ denote the counts of *true positive*, *false positive*, and *false negative* predictions for class k . The macro F_1 score is then derived by macro-averaging the $F_1(k)$ scores across all classes:

$$\text{macro } F_1 = \frac{1}{K} \sum_{k=1}^K F_1(k).$$

This metric provides a comprehensive assessment of the model's performance across classes of varying sizes, balancing precision and recall in the context of class imbalance. In each trial, we selected the model with the highest validation macro F_1 score as the best model, and evaluated its generalization performance on the test set using the macro F_1 score.

C. COMPARISON WITH SSL APPROACHES

First, we demonstrated the effectiveness of our framework by comparing its performance against two popular SSL methods in computer vision, FixMatch [10] and Adsh [21], as well as two recent SSL methods specifically developed for WBM defect pattern classification, SSL-Ensemble [24] and SSL-Dual-Head [14]. Table 3 presents the classification performance in terms of the macro F_1 score, averaged over ten trials with different random seeds. Each column corresponds to a different proportion of labeled training data, and each row to a different training method. Overall, our method consistently outperformed all other methods across all proportions of labeled data. Notably, in the 1% and 5% labeled data scenarios—both realistic from a labeling cost perspective and indicative of model performance in data-scarce environments—our method surpassed the second-best method by 2.0% and 1.7%, respectively.

D. CHOICE OF HYPERPARAMETERS

In Table 4, we evaluate the sensitivity of our method to the momentum coefficient λ , which controls the stability of threshold updates. We report the Macro F_1 score across different choices of $\lambda \in \{0.1, 0.2, 0.5, 0.8, 0.9\}$. The results indicate that ClimEx is highly robust to variations in this hyperparameter. As shown in the table, the performance differences across the tested values are minimal, particularly in the range of [0.5, 0.9]. Even with limited labeled data

(e.g., 1%), the standard errors remain low, suggesting that the momentum mechanism effectively stabilizes training without requiring extensive hyperparameter search. The results show that $\lambda = 0.8$ achieves the highest performance. Therefore, unless otherwise stated, we default to $\lambda = 0.8$ in all experiments.

We conducted an additional experiment to identify the optimal combinations for weak and strong augmentations, $\alpha(\cdot)$ and $\mathcal{A}(\cdot)$, that yield the highest classification performance. To isolate the impact of different augmentations, we used FixMatch as a testbed due to its use of a unified fixed threshold for pseudo-labeling across classes—a feature typical of SSL algorithms designed for balanced class settings. In contrast, Adsh and our method use dynamic per-class thresholds, which we believed could complicate the proper evaluation of the augmentation effects. As shown in Table 5, *rotation* and *flip* were selected as weak augmentations to support reliable pseudo-labeling by maintaining the semantics of the WBM input (e.g., defect orientation invariance). The candidates for strong augmentation were *random select*, *cutout + noise*, and *cutout + crop*. These strong augmentations were chosen to simulate real-world process imperfections and force the model to learn robust global features rather than overfitting to local artifacts. The *random select* option randomly chooses two augmentations from *cropping*, *cutout*, *noise addition*, and *shifting* [12]. Based on their consistent and stable performance across experiments, *rotation* and *cutout + noise* were identified as the most effective weak and strong augmentations, respectively. In all experiments, these were used for FixMatch, Adsh, and our proposed method.

E. COMPARISON WITH A SELF-SUPERVISED CONTRASTIVE LEARNING APPROACH

Comparing SSL and self-supervised contrastive learning (SSCL) is crucial because both aim to leverage unlabeled data to overcome the limitations of scarce labeled data, though they do so through different training frameworks [28], [32], [33], [34]. SSL seeks to enhance classification performance by simultaneously using labeled and unlabeled data in a single-stage learning process. In contrast, SSCL focuses on learning generalizable representations from unlabeled data during a pre-training phase, which then improves downstream classification tasks during fine-tuning. This comparison allows for evaluating the effectiveness of each approach across varying levels of labeled data availability.

We selected WaPIRL [12], a state-of-the-art self-supervised contrastive learning method for WBM defect pattern classification, as our direct competitor. The training configurations differed slightly from those described in Section IV-B. To ensure a fair comparison, experiments were conducted using a VGG16 backbone [35] instead of ResNet-18. Our model was trained for 102,400 iterations with a batch size of 256 to ensure convergence. We used the stochastic gradient descent (SGD) optimizer with a momentum factor of 0.9, a base learning rate of 0.001, and

TABLE 3. Comparison of classification performance using the macro F_1 score (CNN backbone: ResNet-18). For each labeled data proportion, the best macro F_1 score across methods is boldfaced (columnwise). The mean and standard errors (in parentheses) are calculated across ten random trials.

Training Method	Labeled Data					
	1,383 (1%)	6,918 (5%)	13,836 (10%)	34,590 (25%)	69,180 (50%)	138,630 (100%)
FixMatch	0.705	0.803	0.834	0.856	0.870	0.879
[10]	(0.021)	(0.015)	(0.015)	(0.017)	(0.008)	(0.005)
Adsh	0.707	0.806	0.833	0.853	0.872	0.881
[21]	(0.042)	(0.018)	(0.011)	(0.011)	(0.007)	(0.004)
SSL-Ensemble	0.016	0.140	0.203	0.264	0.420	0.478
[24]	(0.035)	(0.048)	(0.094)	(0.143)	(0.068)	(0.045)
SSL-Dual-Head	0.248	0.689	0.795	0.855	0.872	0.887
[14]	(0.241)	(0.115)	(0.024)	(0.005)	(0.006)	(0.006)
ClimEx	0.727	0.823	0.842	0.864	0.876	0.891
(Proposed)	(0.020)	(0.015)	(0.007)	(0.008)	(0.005)	(0.003)

TABLE 4. Evaluation of ClimEx with different momentum coefficients λ . We reported the mean and standard errors across ten independent runs.

λ	Labeled Data					
	1,383 (1%)	6,918 (5%)	13,836 (10%)	34,590 (25%)	69,180 (50%)	138,630 (100%)
0.1	0.716	0.809	0.841	0.861	0.873	0.889
	(0.027)	(0.014)	(0.010)	(0.011)	(0.008)	(0.003)
0.2	0.715	0.811	0.840	0.862	0.875	0.885
	(0.029)	(0.020)	(0.007)	(0.004)	(0.006)	(0.004)
0.5	0.725	0.812	0.838	0.861	0.876	0.887
	(0.026)	(0.009)	(0.005)	(0.007)	(0.004)	(0.003)
0.8	0.727	0.823	0.842	0.864	0.876	0.891
	(0.020)	(0.015)	(0.007)	(0.008)	(0.005)	(0.003)
0.9	0.721	0.812	0.842	0.862	0.874	0.886
	(0.020)	(0.011)	(0.014)	(0.006)	(0.007)	(0.004)

TABLE 5. Experimental results to determine the baseline combinations of weak and strong augmentations using macro F_1 score. We reported the mean and standard errors calculated across ten independent runs.

Training Method	Weak Augmentation	Strong Augmentation	Label Data					
			1,383 (1%)	6,918 (5%)	13,836 (10%)	34,590 (25%)	69,180 (50%)	138,630 (100%)
FixMatch	Rotation	Cutout + Noise	0.705 (0.021)	0.803 (0.015)	0.834 (0.015)	0.856 (0.017)	0.870 (0.008)	0.879 (0.005)
		Cutout + Crop	0.689 (0.026)	0.778 (0.019)	0.809 (0.020)	0.831 (0.018)	0.845 (0.013)	0.862 (0.009)
		Random select	0.608 (0.033)	0.734 (0.028)	0.749 (0.030)	0.808 (0.022)	0.818 (0.014)	0.831 (0.011)
	Flip	Cutout + Noise	0.641 (0.027)	0.773 (0.022)	0.794 (0.018)	0.823 (0.019)	0.839 (0.011)	0.854 (0.005)
		Cutout + Crop	0.633 (0.031)	0.758 (0.020)	0.780 (0.019)	0.812 (0.015)	0.826 (0.016)	0.851 (0.010)
		Random select	0.629 (0.033)	0.713 (0.037)	0.733 (0.025)	0.782 (0.019)	0.804 (0.015)	0.829 (0.013)

a weight decay factor of 0.00001. The learning rate was adjusted using a half cosine schedule [36]. A grid search over the values {0.01, 0.005, 0.001, 0.0005, 0.00001} was conducted to determine the optimal learning rate and weight decay. Model selection was based on the macro F_1 score, measured on the validation set every 512 iterations. The test macro F_1 scores, summarized in Table 6, show that our method is on par with WaPRL across all proportions of labeled data. Notably, unlike SSCL approaches that require two distinct phases—unsupervised pre-training followed by

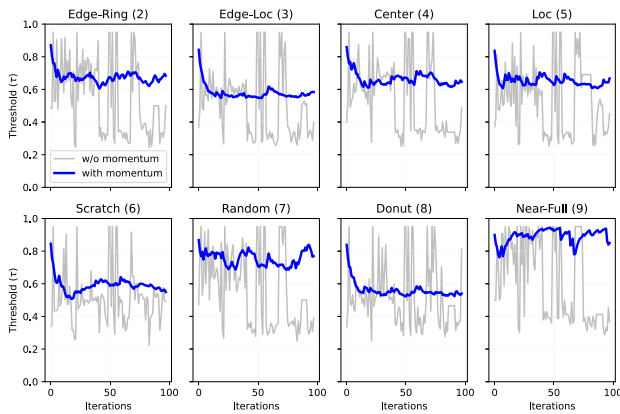
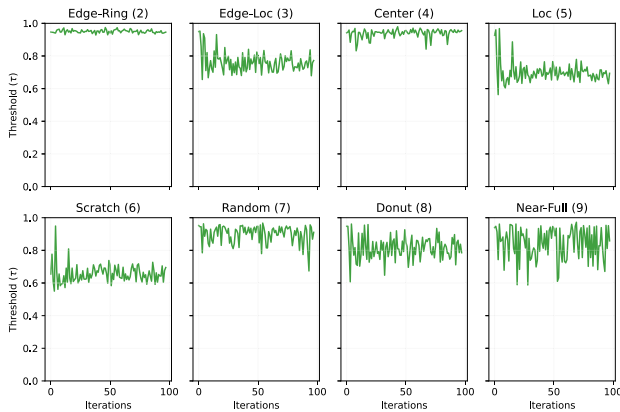
supervised fine-tuning—our method unifies these stages into a single phase. This integration allows for consideration of class distribution during model training, significantly enhancing our model's ability to effectively address class imbalance.

V. DISCUSSION

As previously noted, the momentum-based update mechanism in Eq. (10) is crucial for avoiding undesirable fluctuations in per-class thresholds under an extreme class

TABLE 6. Comparison with a self-supervised learning approach using macro F_1 scores. The best macro F_1 score across methods and labeled data proportions is boldfaced for each column. We reported the mean and standard errors across ten independent runs.

Encoder Architecture	Training Method	Weak Augmentation	Strong Augmentation	Labeled Data					
				1,383 (1%)	6,918 (5%)	13,836 (10%)	34,590 (25%)	69,180 (50%)	138,630 (100%)
VGG16	WaPIRL [12]		Crop	0.741	0.815	0.839	0.864	0.880	0.897
	ClimEx (proposed)	Rotation	Cutout + Noise	0.743 (0.006)	0.820 (0.004)	0.855 (0.011)	0.867 (0.007)	0.884 (0.003)	0.901 (0.002)

**FIGURE 2.** Per-class thresholds plotted as a function of iterations, with (blue) and without (gray) the momentum-based update mechanism. Both models were trained with 1% labeled data.**FIGURE 3.** Per-class thresholds (without momentum) plotted as a function of iterations, under an simulated scenario with a class imbalance ratio of 100 between the majority class 'none', and the minority class 'near-full'.

imbalance setting, as observed in the WM-811K benchmark. Preliminary experiments indicated that without this mechanism, model training becomes unstable and prone to suboptimal solutions. We trained two models, with and without momentum, to explore how the thresholds change over time. In Figure 2, we plotted the threshold values as a function of iterations, for all classes except the majority class, 'None'. Without momentum-based updates (gray), the thresholds fluctuate substantially, which is undesirable because it can lead to erroneous pseudo-labels hampering

performance at different stages of training. We believe it is advantageous for the thresholds to decrease almost monotonically as training progresses, which is the case when the momentum-based update mechanism is applied (blue). This is a valid statement, as in the early stages of training, when the model is likely underfitting, higher thresholds should allow only examples with highly confident predictions to be pseudo-labeled. On the other hand, pseudo-labels with lower confidence should be introduced only after the model has established clearer decision boundaries.

To further investigate the question of when the thresholds fluctuate, we conducted an additional simulation under a less extreme class imbalance setup, similar to the setting discussed in Adsh [21]. First, we randomly divided the original labeled data into a 30% labeled and a 70% unlabeled dataset, completely discarding the original unlabeled set. This approach allowed us to maintain full control over the imbalance ratio, given the labels for every data instance. Next, we simulated an imbalance ratio of 100 in which the sample size of the majority class, 'None', is reduced to 100 times the size of the minority class, 'Near-Full'. As shown in Figure 3, the thresholds are relatively stabilized without the momentum-based update mechanism, in turn supporting our statement that the fluctuations were primarily driven by the extreme level of class imbalance. This suggests that the momentum-based update mechanism is an essential component of the proposed method for handling extreme class imbalance.

VI. CONCLUSION

In this study, we addressed the significant challenges of small labeled datasets and imbalanced class distributions in WBM defect pattern classification. We introduced ClimEx, a novel semi-supervised learning framework based on consistency regularization and pseudo-labeling, which effectively mitigates the impact of class imbalance through dynamically updated, class-specific thresholds for pseudo-labeling. Using a publicly available dataset from a real-world fabrication process, we demonstrated the effectiveness of our method from multiple perspectives, showing its potential to enhance WBM defect pattern classification in real-world, class-imbalanced environments.

For the semiconductor industry, the proposed method offers a promising, automated, and reliable solution for wafer quality assessment. By leveraging both labeled and unlabeled data and effectively managing inherent class imbalance, our approach can assist manufacturers in identifying process

failures and their root causes, leading to improved yields and reduced production costs. ClimEx opens up new possibilities for more accurate and dependable WBM defect pattern classification in semiconductor manufacturing.

Although ClimEx has shown promising results in overcoming key obstacles in WBM defect pattern classification, there are opportunities for further research to enhance this approach. Future work could focus on systematically selecting the most effective augmentation strategies for consistency regularization. Exploring a broader range of data augmentation techniques and employing automated methods, such as Bayesian optimization or reinforcement learning-based strategies, could efficiently optimize the augmentation process and improve performance. Additionally, incorporating out-of-distribution detection capabilities into ClimEx presents a promising direction for future research. As new fabrication processes emerge or existing ones evolve, novel defect patterns may appear. Enhancing the model's ability to detect out-of-distribution examples would ensure the system's robustness and accuracy in identifying unforeseen defects. Furthermore, the momentum-based thresholding mechanism in ClimEx is fundamentally designed to handle extreme imbalance, which may offer a methodological basis for extension to other industrial datasets or cross-fab scenarios with similar data characteristics. Advancements in these directions will significantly contribute to the semiconductor industry's progress, empowering it with accurate and automated wafer quality assessment and failure analysis capabilities. We expect the proposed method to lay a strong foundation for future developments in WBM defect pattern classification and hold the potential to revolutionize defect inspection systems in semiconductor manufacturing processes.

REFERENCES

- [1] K. Kyeong and H. Kim, "Classification of mixed-type defect patterns in wafer bin maps using convolutional neural networks," *IEEE Trans. Semicond. Manuf.*, vol. 31, no. 3, pp. 395–402, Aug. 2018.
- [2] Y. Bae and S. Kang, "Supervised contrastive learning for wafer map pattern classification," *Eng. Appl. Artif. Intell.*, vol. 126, Nov. 2023, Art. no. 107154.
- [3] M.-J. Wu, J.-S.-R. Jang, and J.-L. Chen, "Wafer map failure pattern recognition and similarity ranking for large-scale data sets," *IEEE Trans. Semicond. Manuf.*, vol. 28, no. 1, pp. 1–12, Feb. 2015.
- [4] D.-H. Lee, "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in *Proc. Workshop challenges Represent. Learn.*, 2013, vol. 3, no. 2, p. 896.
- [5] S. Laine and T. Aila, "Temporal ensembling for semi-supervised learning," in *Proc. Int. Conf. Learn. Represent.*, 2016. [Online]. Available: <https://openreview.net/forum?id=BJ6oOfage>
- [6] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1195–1204.
- [7] D. Berthelot, N. Carlini, I. Goodfellow, N. Papernot, A. Oliver, and C. Raffel, "MixMatch: A holistic approach to semi-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 5050–5060.
- [8] D. Berthelot, N. Carlini, E. D. Cubuk, A. Kurakin, K. Sohn, H. Zhang, and C. Raffel, "ReMixMatch: Semi-supervised learning with distribution matching and augmentation anchoring," in *Proc. Int. Conf. Learn. Represent.*, 2020. [Online]. Available: <https://openreview.net/forum?id=HklkeR4KPB>
- [9] W.-N. Hsu, "Unsupervised data augmentation for consistency training," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 6256–6268.
- [10] K. Sohn, D. Berthelot, N. Carlini, Z. Zhang, H. Zhang, C. A. Raffel, E. D. Cubuk, A. Kurakin, and C.-L. Li, "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 596–608.
- [11] Y. Kong and D. Ni, "A semi-supervised and incremental modeling framework for wafer map classification," *IEEE Trans. Semicond. Manuf.*, vol. 33, no. 1, pp. 62–71, Feb. 2020.
- [12] H. Kahng and S. B. Kim, "Self-supervised representation learning for wafer bin map defect pattern classification," *IEEE Trans. Semicond. Manuf.*, vol. 34, no. 1, pp. 74–86, Feb. 2021.
- [13] H. Hu, C. He, and P. Li, "Semi-supervised wafer map pattern recognition using domain-specific data augmentation and contrastive learning," in *Proc. IEEE Int. Test Conf. (ITC)*, Oct. 2021, pp. 113–122.
- [14] S. Manivannan, "Semi-supervised imbalanced classification of wafer bin map defects using a dual-head CNN," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 122301. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957417423028038>
- [15] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [16] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
- [17] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," Dept. Comput. Sci., Univ. Toronto, Toronto, ON, Canada, Tech. Rep. TR-2009, 2009.
- [18] N. Japkowicz and S. Stephen, "The class imbalance problem: A systematic study," *Intell. data Anal.*, pp. 429–449, 2002.
- [19] J. Kim, Y. Hur, S. Park, E. Yang, S. J. Hwang, and J. Shin, "Distribution aligning refinery of pseudo-label for imbalanced semi-supervised learning," *Adv. Neural Inf. Process. Syst.*, pp. 14567–14579, Dec. 2020.
- [20] C. Wei, K. Sohn, C. Mellina, A. Yuille, and F. Yang, "CRcST: A class-rebalancing self-training framework for imbalanced semi-supervised learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10852–10861.
- [21] L.-Z. Guo and Y.-F. Li, "Class-imbalanced semi-supervised learning with adaptive thresholding," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 8082–8094.
- [22] T. Nakazawa and D. V. Kulkarni, "Wafer map defect pattern classification and image retrieval using convolutional neural network," *IEEE Trans. Semicond. Manuf.*, vol. 31, no. 2, pp. 309–314, May 2018.
- [23] M. Saqlain, Q. Abbas, and J. Y. Lee, "A deep convolutional neural network for wafer defect identification on an imbalanced dataset in semiconductor manufacturing processes," *IEEE Trans. Semicond. Manuf.*, vol. 33, no. 3, pp. 436–444, Aug. 2020.
- [24] S. Manivannan, "An ensemble-based deep semi-supervised learning for the classification of wafer bin maps defect patterns," *Comput. Ind. Eng.*, vol. 172, Oct. 2022, Art. no. 108614. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0360835222006040>
- [25] Y. Yang and Z. Xu, "Rethinking the value of labels for improving class-imbalanced learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 19290–19301.
- [26] A. Oliver, A. Odena, C. Raffel, E. D. Cubuk, and I. Goodfellow, "Realistic evaluation of deep semi-supervised learning algorithms," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 3239–3250.
- [27] O. J. Hénaff, A. Srinivas, J. D. Fauw, A. Razavi, C. Doersch, S. M. A. Eslami, and A. V. D. Oord, "Data-efficient image recognition with contrastive predictive coding," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 4182–4192.
- [28] T. Chen, S. Kornblith, K. Swersky, M. Norouzi, and G. E. Hinton, "Big self-supervised models are strong semi-supervised learners," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 22243–22255.
- [29] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
- [30] N. Srivastava, G. E. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.
- [31] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2014.

- [32] X. Zhai, A. Oliver, A. Kolesnikov, and L. Beyer, "S4L: Self-supervised semi-supervised learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1476–1485.
- [33] T. Chen, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, vol. 1, 2024, pp. 1597–1607.
- [34] M. Zheng, S. You, L. Huang, F. Wang, C. Qian, and C. Xu, "SimMatch: Semi-supervised learning with similarity matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14451–14461.
- [35] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
- [36] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. Int. Conf. Learn. Represent.*, 2016. [Online]. Available: <https://openreview.net/forum?id=Skq89Scxx>



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